

ECOCIVIX AI — MASTER PRODUCT REQUIREMENTS DOCUMENT

Document: 01 — Master PRD

Status: Pre-Hackathon Planning

Version: 0.1

Product: ECOCIVIX AI

Working Tagline: Safer. Smarter. More Sustainable Communities.

1. Product Vision

ECOCIVIX AI is an AI-powered community intelligence platform designed to help citizens report civic, environmental, and community-safety problems and transform those reports into structured, prioritized, actionable information.

The central idea:

Citizen observes a problem → reports it → AI understands it → impact is assessed → issue is routed → action is tracked → community intelligence emerges.

ECOCIVIX combines three connected intelligence layers:

1. **Eco Intelligence** — understands environmental and sustainability-related problems.
2. **Civic Intelligence** — understands, prioritizes, and routes civic problems.
3. **Community Safety** — identifies safety-related concerns and provides basic emergency-support workflows.

The product should feel like one coherent platform rather than three separate applications.

2. Problem Statement

Communities face interconnected civic, environmental, and safety problems such as waste accumulation, water leakage, pollution, damaged infrastructure, poor sanitation, broken streetlights, and public-safety concerns.

These problems are often reported through disconnected channels, making it difficult to understand urgency, identify the appropriate department, recognize recurring patterns, and coordinate action.

ECOCIVIX AI aims to turn individual citizen reports into actionable community intelligence by automatically understanding the problem, assessing relevant environmental and safety impacts, prioritizing it, and helping route it toward appropriate action.

Short Problem Statement

Communities struggle with civic, environmental, and safety issues that are often reported but poorly prioritized and coordinated.

ECOCIVIX AI uses AI to understand these problems, assess their impact, and turn them into actionable community solutions.

3. Product Objectives

Primary Objectives

- Make reporting community problems simple.
- Reduce manual classification after a report.
- Automatically identify issue category and priority.
- Identify environmental impact where relevant.
- Identify safety implications where relevant.
- Suggest the responsible department or action area.
- Allow citizens to track issue progress.
- Give administrators an actionable issue-management dashboard.
- Surface recurring issues and community hotspots.
- Provide a polished, understandable, trustworthy experience.

Secondary Objectives

- Demonstrate practical AI rather than AI used only as a chatbot.
- Connect civic problems with environmental and safety consequences.
- Support data-driven community decision-making.
- Create a foundation for broader smart-community workflows.

4. Target Users

Citizens

Citizens can report civic/environmental/safety issues, add descriptions and optional images, provide location, receive AI analysis, track status, and access safety-support functionality where relevant.

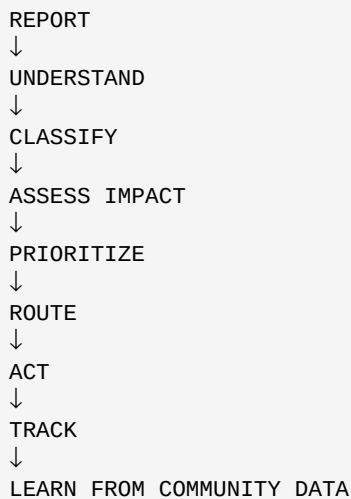
Administrators / Authorities

Administrators can view, search, filter, assign, update, resolve, and close issues and inspect AI analysis.

Community Intelligence View

A future-facing intelligence layer can aggregate anonymized issue patterns into issue density, category distribution, environmental hotspots, safety-risk clusters, recurring infrastructure problems, and resolution trends.

5. Core Product Loop



6. Core Feature Set

Citizen Reporting

A report can contain:

- Title
- Description
- Optional image
- Location
- Issue context

Examples: garbage dumping, water leakage, broken streetlight, road damage, overflowing waste bin, pollution concern, unsafe public area.

AI Issue Understanding

AI can determine:

- Category
- Priority
- Short summary
- Responsible department
- Environmental impact
- Safety relevance/risk where appropriate
- Suggested action

AI output must be structured and validated.

Issue Tracking

Citizens can see:

- Current status
- Issue details
- AI analysis
- Timeline/activity
- Administrative updates
- Resolution state

Admin Operations

Administrators can:

- Search issues.
- Filter issues.
- Open issue details.
- Review AI analysis.
- Assign issues.
- Change status.
- Add updates.
- Resolve/close issues.

Community Intelligence

The platform can aggregate reports into category distribution, priority distribution, environmental-impact patterns, safety-related clusters, geographic hotspots, and recurring issue patterns.

This is intended as an intelligence layer, not merely a statistics page.

7. Three Intelligence Pillars

Eco Intelligence

Relevant issues include waste, water leakage/wastage, pollution, cleanliness, energy wastage, and environmental hazards.

Possible outputs:

- Environmental relevance.
- Impact level: Low / Medium / High.
- Affected resource or environmental area.
- Suggested sustainable action.

Avoid unsupported claims such as exact CO₂ savings unless a defensible calculation method and data are available.

Civic Intelligence

Relevant issues include roads, streetlights, public infrastructure, water/electricity problems, sanitation, and public facilities.

AI should help determine category, priority, responsible department, and suggested action.

Community Safety

The initial safety layer should remain focused and realistic:

- Safety incident reporting.
- Safety-related civic issue reporting.
- Emergency-support workflow.
- Emergency contact support.
- Location sharing with chosen contacts.
- Identification of civic conditions that may create safety concerns.

Safety features are assistance tools and must not imply guaranteed protection or replace emergency services.

8. Cross-Domain Intelligence

One issue can have multiple dimensions.

Example:

“Streetlight broken near girls' hostel.”

Potential AI interpretation:

- Civic category: Public Infrastructure / Streetlight
- Priority: High
- Safety relevance: High
- Environmental impact: Low
- Responsible area: Electrical Maintenance
- Suggested action: Inspect and repair lighting
- Location: Reported location

This demonstrates how ECOCIVIX connects civic intelligence with community safety.

9. AI Architecture — Product-Level Requirement

Input

- User-provided title.
- User-provided description.
- Optional image.
- Location/context where available.

Processing

1. Validate input.
2. Understand issue.
3. Classify category.
4. Determine priority.
5. Generate concise summary.
6. Determine department/action area.
7. Assess environmental relevance.
8. Assess safety relevance where applicable.
9. Generate suggested action.
10. Validate structured output.
11. Return results.

Conceptual Output

```
category
priority
summary
department
environmentalImpact
environmentalReason
safetyRisk
safetyReason
suggestedAction
```

The exact implementation contract will be finalized in the AI Intelligence Specification and API Contract.

10. AI Reliability Requirements

- Validate allowed categories.
- Validate allowed priority values.
- Require non-empty summaries.
- Validate structured output.
- Handle malformed responses.
- Handle AI timeout/failure.
- Provide a clearly defined fallback where appropriate.
- Never claim a fallback result was generated by the AI provider.
- Never expose API keys in frontend code.
- Keep secrets server-side.

11. Issue Lifecycle

Conceptual lifecycle:

```
OPEN → ASSIGNED → IN_PROGRESS → RESOLVED → CLOSED
```

A report may also be rejected where appropriate.

Final transition rules will be defined in the technical/API specification.

12. Trust, Safety & Privacy Principles

ECOCIVIX may handle locations, reports, images, and safety-related information.

The product should:

- Minimize unnecessary personal data.
- Protect authentication credentials.
- Keep secrets server-side.
- Restrict administrative actions.
- Validate user input.
- Use secure authentication.
- Avoid exposing private issue information unnecessarily.
- Clearly distinguish AI-generated analysis from verified human decisions.
- Avoid presenting AI classifications as absolute facts.
- Treat emergency functionality as support rather than guaranteed protection.

13. UX Principles

Clear

A user should understand what to do without documentation.

Fast

Reporting should require minimal friction.

Intelligent

AI should reduce effort rather than add complexity.

Trustworthy

Users should understand why an issue received a classification or priority.

Premium

The interface should feel polished and intentional rather than like a generic admin dashboard.

Accessible

Important actions, status information, and emergency-related controls must remain visually clear.

14. Visual Product Direction

The intended visual language is:

- Premium.
- Modern.

- Futuristic but usable.
- Clean.
- Intelligent.
- Environmental.
- Civic.
- High-quality SaaS.
- Strong visual hierarchy.
- Purposeful motion.

3D and animation should support storytelling and hierarchy rather than become decoration.

Detailed visual rules will be defined in the Frontend UI/UX Bible, 3D Experience & Animation Specification, and Design System.

15. Proposed Major Screens

Public

- Landing page.
- Product overview.
- How ECOCIVIX works.
- Sign in.
- Sign up.

Citizen

- Citizen dashboard.
- Report issue.
- AI analysis result.
- My issues.
- Issue details.
- Issue timeline.
- Community intelligence/map view.
- Safety center.

Admin

- Admin dashboard.

- Issue management.
- Issue detail.
- Assignment.
- Status management.
- Activity/update timeline.
- Community intelligence.
- Hotspot/map view.

16. Dashboard Intelligence

Potential indicators:

- Open issues.
- High-priority issues.
- Issues requiring attention.
- Recently resolved issues.
- Environmental issue distribution.
- Safety-related issue distribution.
- Geographic clusters.

Any metric displayed in the final product must be backed by actual application data. No fake statistics in the final demo.

17. Community Intelligence Map

Potential capabilities:

- Issue markers.
- Category filtering.
- Priority filtering.
- Environmental-impact filtering.
- Safety-risk filtering.
- Clustered issue visualization.
- Hotspot identification.

A hotspot should be derived from actual report data and a documented aggregation method.

18. Non-Goals for the Initial Hackathon Build

Do not attempt:

- Advanced real-time surveillance.
- Facial recognition.
- Crime prediction.
- Guaranteed emergency response.
- Complex government integrations unless explicitly available.
- Unsupported environmental calculations.
- Excessive AI agents with unclear value.
- Dozens of unrelated features.
- A social-media platform.
- A generic AI chatbot disconnected from the core workflow.

19. Hackathon MVP

Citizen

1. Authentication.
2. Report creation.
3. Image/location where feasible.
4. AI analysis.
5. Issue tracking.
6. Status/timeline.

AI

1. Category.
2. Priority.
3. Summary.
4. Department.
5. Environmental impact.
6. Safety relevance.
7. Suggested action.

Admin

1. Issue dashboard.

2. Search/filter.
3. Issue details.
4. Assignment.
5. Status updates.
6. Resolution workflow.

Intelligence

1. Issue distribution.
2. Community hotspots/map.
3. Environmental/safety insights.

20. Product Success Criteria

A judge should be able to understand and experience the core loop without manual intervention:

```
Create Report
↓
AI Analysis
↓
Issue Appears in Admin
↓
Admin Acts
↓
Citizen Sees Update
↓
Community Insight Updates
```

The demonstrated flow should use real application data.

21. Technical Direction

Preferred direction:

- Frontend: React + Vite + JavaScript + CSS
- Backend: Node.js + Express + JavaScript
- Database: Supabase/PostgreSQL
- Authentication: Supabase Auth
- Storage: Supabase Storage where required
- AI: Gemini/free-tier provider where available
- Maps/location: suitable free/open solution where practical
- Deployment: free/low-cost services where available

Exact choices remain subject to official hackathon requirements and final technical architecture.

22. Development Philosophy

Build* → *Integrate* → *Test* → *Polish* → *Freeze

Priority order:

1. Core functionality.
2. Real integration.
3. Reliability.
4. Security.
5. Testing.
6. UX quality.
7. Performance.
8. Visual polish.
9. Optional features.

If integration is broken, fix integration before adding features.

23. AI-Assisted Team Workflow

```
Garv – Final Product Authority
↓
DeepSeek – AI Project Manager
↓
AI Tool Assignment
↓
Antigravity – Builder
↓
Testing / Review
↓
Fix → Retest Loop
↓
Garv – Final Approval
```

DeepSeek plans, breaks down work, prepares implementation prompts, analyzes results, and suggests debugging. It cannot approve phases, change architecture without Garv, merge into main, force-push, rewrite history, declare completion without verification, or override Garv.

Antigravity is the primary implementation agent: inspect, implement, test, fix, and report.

Garv is final authority for architecture, scope, feature approval, integration, phase approval, security-sensitive decisions, and final demo readiness.

24. Hackathon Execution Principle

Avoid feature explosion.

If the core product works:

Improve reliability and presentation before adding another feature.

If the core product does not work:

Stop adding features and fix the core flow.

Feature freeze should happen before final presentation/rehearsal.

25. Future Scope

Potential expansion:

- Advanced environmental analytics.
- Larger community datasets.
- Authority integrations.
- Automated recurring issue detection.
- Multilingual reporting.
- Image-based issue analysis.
- Advanced geospatial intelligence.
- Community participation mechanisms.
- Sustainability recommendations.
- Public transparency dashboards.

Future scope must not distract from the hackathon MVP.

26. Product North Star

ECOCIVIX should answer five questions for every meaningful community problem:

What is happening?

How important is it?

Who should act?

What impact does it have?

What should happen next?

That is the core intelligence behind the product.

27. Document Dependencies

This Master PRD is the source document for:

- 02_FRONTEND_UI_UX_BIBLE.md
- 03_3D_EXPERIENCE_ANIMATION_SPEC.md
- 04_DESIGN_SYSTEM.md
- 05_AI_INTELLIGENCE_SPEC.md
- 06_TECHNICAL_ARCHITECTURE.md
- 07_DATABASE_DATA_MODEL.md
- 08_API_CONTRACT.md
- 09_SAFETY_TRUST_SPEC.md
- 10_HACKATHON_EXECUTION_PLAN.md
- 11_TESTING_QA_STRATEGY.md
- 12_DEMO_PRESENTATION_BLUEPRINT.md

Changes to the core product concept should be reflected here first before downstream documents are changed.

END OF MASTER PRD — VERSION 0.1